# Choptuik's echo period and critical exponent at large dimension: evidence for $\Delta_\infty = 1.643893746$ and an $r^2$-shift symmetry behind $\gamma_\infty = 1/2$

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We compute the echo period $\Delta(D)$ of Choptuik's critical solution of massless scalar collapse in $D$ dimensions at 857 points $8.25 \le D \le 61.03$ with a double-double (about 32-digit) solver, and the critical exponent $\gamma$ of the black-hole radius at 67 points up to $D = 61$; an independent double-precision solver covers $3.6 \le D \le 40$. One rational function of $1/(D-2)$ reproduces both for all $D \ge 4$ and agrees with the published values; a $6 \times 10^{-5}$ difference at $D = 5$ is traced to resolution in the published data. Held-out tests support, but do not prove, analyticity in $1/D$, which gives $\Delta_\infty = 1.643893746 \pm 1.5 \times 10^{-8}$; an approach to $\Delta = 0$, as conjectured in the literature, is excluded by the data if it is a power law (exponent at least 0.01) and disfavoured if it is logarithmic. The exponent tends to $\gamma_\infty = 1/2$ to about $5 \times 10^{-7}$. Within the measured large-$D$ scaling the $1/D$ expansion is unobstructed for every period, so $\Delta$ and $\gamma$ are selected beyond all orders in $1/D$, at radii of order $\sqrt{D}$ times the time to the singularity, where the field equations are Rozali and Way's nonlinear heat equation. Its shift symmetry in $r^2$ gives an exact unstable mode with $1/\gamma = 2$ at $D = \infty$, which explains $\gamma_\infty = 1/2$ given a conjectured selection rule, the uniqueness of the unstable mode and continuity in $1/D$; that rule, evaluated on the $D = \infty$ equations alone, gives $\Delta_\infty \approx 1.644$ and the shape of the echo. For the axion–dilaton, whose large-$D$ equations keep the symmetry, $\gamma \to 1/2$ (computed to $D = 80$); for matter with a centrifugal term, which breaks it, the computations so far do not decide $\gamma_\infty$. The work was done by an AI model (Claude, Anthropic) under the author's direction and has not been reviewed by a human expert (Sec. VIII).

## I. INTRODUCTION

Fine-tuning a one-parameter family of initial data for a massless scalar field to the threshold of black-hole formation, Choptuik found that the evolution approaches a universal critical solution that repeats itself on ever smaller scales: it is discretely self-similar (DSS) with echo period $\Delta$, and black holes formed just above threshold have a size that scales as a power of $p - p^*$ [1]. The scaling follows from the critical solution having a single unstable mode, with exponent $\gamma = 1/\lambda$ for its growth rate $\lambda$ [2]. Throughout, $\gamma$ is the *radius* exponent, $r_{\rm BH} \propto (p - p^*)^\gamma$; in $D$ spacetime dimensions the mass then scales as $M \propto (p - p^*)^{(D-3)\gamma}$, and in four dimensions the two exponents coincide. In four dimensions $\Delta = 3.445452402(3)$ [3] and $\gamma \simeq 0.374$ [4]; see the review [5].

The dependence on the spacetime dimension $D$ has been studied numerically, mostly at percent-level precision, for $3.5 \le D \le 14$ (mostly integer $D$) [6–11], and to about $10^{-6}$ for continuous $3.05 < D < 5.5$ by Ecker, Ecker, Grumiller and Jechtl [12], who find a maximum of $\Delta$ near $D = 3.756$. Large $D$ has been a fruitful limit in gravity [13]. For critical collapse, Rozali and Way [14] derived the leading-order large-$D$ equations of a self-gravitating scalar field, observed that they are parabolic with the roles of space and time swapped, and estimated $\gamma = 1/2$ from the divergence of their oscillating solutions. Clark and Pimentel [15] constructed continuously self-similar (CSS) critical solutions at large $D$ by matched asymptotics. Ecker, Ecker and Grumiller [16] constructed analytic DSS solutions at large $D$, found that at leading order any periodic profile with any period solves the equations, and, with [12], conjectured $\Delta \to 0$ as $D \to \infty$; they also report the expectation $\gamma \to 1/2$. Wu [17] studied a quantum growing mode of the collapsing matter in continuously and discretely self-similar critical spacetimes at large $D$. Bland's thesis [10] hypothesised $\Delta \to 5/2$.

This paper computes $\Delta(D)$ and $\gamma(D)$ to high precision up to $D = 61$, asks what happens as $D \to \infty$, and explains what it can from the structure of the equations at large $D$. It reports:

1. $\Delta(D)$ at 857 points $8.25 \le D \le 61.03$ from a double-double (about 32-digit) solver, $\gamma(D)$ at 67 points up to $D = 61$, and both for $3.6 \le D \le 40$ from an independent double-precision solver; one rational function of $1/(D-2)$ that reproduces them for all $D \ge 4$; and the comparison with published values, including a $6 \times 10^{-5}$ difference at $D = 5$ that the published higher-resolution data trace to resolution (Sec. III).

2. Held-out tests that support analyticity of $\Delta$ in $1/D$, giving $\Delta_\infty = 1.643893746 \pm 1.5 \times 10^{-8}$; a power-law approach to $\Delta = 0$ (exponent at least 0.01) is excluded and a logarithmic one disfavoured by the data to

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$D = 61$; given analyticity, nine constants tested as closed forms for $\Delta_\infty$ are excluded; and $\gamma_\infty = 1/2$ to about $5 \times 10^{-7}$ (Sec. IV).

3. A physical picture of the large-$D$ limit (Sec. V). Inside the light cone the echo becomes a weak free wave, which on its own cannot echo; the period is fixed by effects smaller than every power of $1/D$, in an inner region at radii of order $\sqrt{D}$ times the time to the singularity, where the equations become Rozali and Way's nonlinear heat equation. Its shift symmetry in $r^2$ gives an exact unstable mode with $\lambda = 2$, which explains $\gamma_\infty = 1/2$ under stated conditions, and a conjectured selection rule posed there reproduces $\Delta_\infty$ to about $10^{-4}$ and the shape of the echo from the $D = \infty$ equations alone.

4. Tests on other matter (Sec. VI): the axion–dilaton and the complex scalar keep the $r^2$-shift at large $D$; for the axion–dilaton $\gamma \to 1/2$, and for both the selection rule reproduces the limiting frequency. For models with a centrifugal term, which breaks the symmetry, the computations so far do not decide $\gamma_\infty$.

The theory behind item 3 (the large-$D$ expansion and its Stokes structure, the selection rule with its hypotheses, and a computer-assisted proof in a model case) is developed in a companion paper [18]; here we state its results in physical terms, with the numerical evidence, and refer there for proofs. We assume, without checking it at large $D$, that the branch we follow remains the critical solution, with a single unstable mode (Sec. VII).

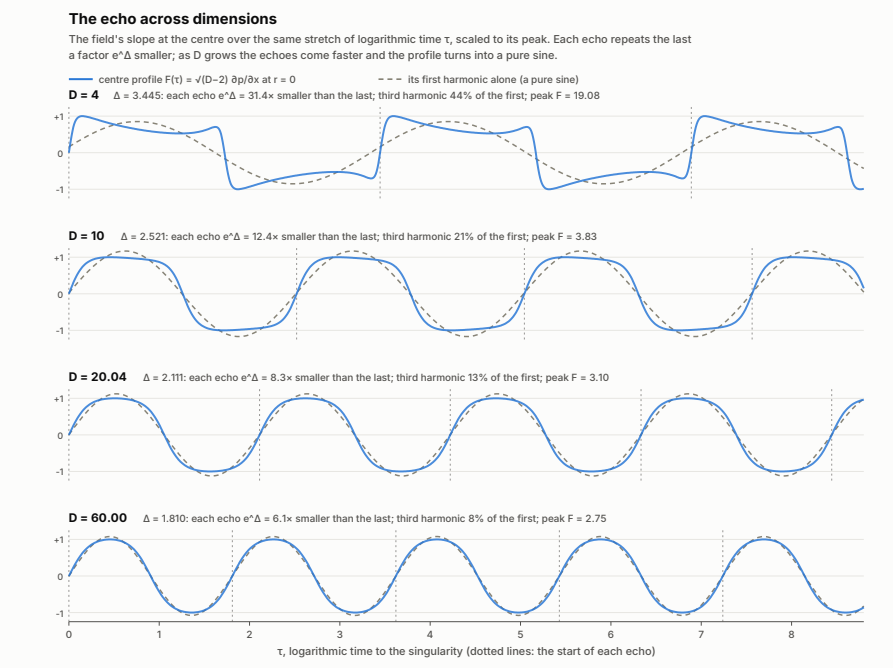



FIG. 1. The echo across dimensions: the centre slope of the field, $F(\tau) = \sqrt{D-2}\,\partial_x p$ at $r = 0$, over the same stretch of logarithmic time $\tau$ for $D = 4$, 10, 20 and 60, each scaled to its peak (dashed: its first harmonic). At $D = 4$ the profile is close to a square wave and each echo is $e^{\Delta} = 31\times$ smaller than the last; at $D = 60$ it is nearly a pure sine and the factor is 6.1. The $D = 4$ profile is from the double-precision solver (39 harmonics); the others are double-double solutions.

## II. EQUATIONS AND NUMERICAL METHOD

We use the self-similar radius $x$ (radius over time to the singularity, normalized so that the past light cone of the singularity is at $x = 1$) and $\tau = -\ln(-t)$, or the phase $\theta = \omega\tau$ with $\omega = 2\pi/\Delta$. With $n = D - 2$, $p$ and $m$ the (scaled) derivatives of the scalar field along the two radial null directions, $a$ the radial metric function ($b = a^2$) and $c$ the lapse-type function that sets the light cones, the equations are

$$x c_x = (n-1)\,c\,(b-1), \qquad x b_x = b\,(p^2 + m^2) - (n-1)\,b\,(b-1), \tag{1}$$

$$x(x-c)\,p_x = -\omega x\,p_\theta + Gp - cp + \tfrac{n}{2}\,c\,(p+m), \qquad x(x+c)\,m_x = -\omega x\,m_\theta - Gm + cm - \tfrac{n}{2}\,c\,(p+m), \tag{2}$$

with $G = (n-1)\,c\,(b-1)$, on $0 \le x \le 1$; $p$, $m$ contain only odd harmonics of $\theta$ and $b$, $c$ only even ones; the solution is regular at $x = 0$, analytic at the past light cone $x = 1$, and $c(1) = 1$ fixes the gauge: a periodic boundary-value problem with $\Delta$ as an eigenvalue [19].

*Double-precision solver* ($3.6 \le D \le 40$): spectral multiple shooting [39] in $x$ with Fourier modes in $\tau$ (23–47 modes), with the integration-step error extrapolated out; near $D = 4$ the points are extrapolated in the mode count. Above $D \approx 26$ rounding at $10^{-16}$ in the residual moves $\Delta$ by $10^{-7}$–$10^{-6}$ at $D = 30$–$40$.

*Double-double solver* ($8.25 \le D \le 61.03$): double-double arithmetic [41] (about 32 digits) on SIMD vectors; Taylor series in $x$ of order 36 with truncation $10^{-32}$; centre and horizon series to $10^{-33}$; collocation in $\tau$ on the half period (32 points); an exact Jacobian by forward-mode duals [40] with a double-double LU; continuation in $D$ with deflated Newton steps. The Jacobian has a ladder of near-null directions (high $\tau$-harmonics of the centre data, singular values falling roughly like $e^{-0.7k}$ with the harmonic $k$); from $D = 35$ the centre harmonics above 13 are pinned, which moves $\Delta$ by $5 \times 10^{-15}$ at $D = 35$ and about $10^{-15}$ at $D = 40$–$60$, and $\Delta_\infty$ by at most $5.5 \times 10^{-10}$. The checks are listed in Table I. The 857 points fit smooth curves to about $4 \times 10^{-15}$; the estimated error per point is about $10^{-16}$ below $D = 27$ and up to a few $10^{-15}$ above (from the weak directions on 27–35, the pinning above $D = 35$, and the relaxed tolerance above $D = 60.1$). Above $D = 40$ these checks are internal to the double-double solver (resolution, pinning, a second build on a second machine); the independent double-precision solver stops at $D = 40$.

Why the extra precision is needed is itself a physical result. The sensitivity of $\Delta$ to the residual, $\max|\partial\Delta/\partial F|$, grows about 1.8-fold per unit of $D$ below $D = 27$ and 1.5-fold at $D = 56$–$61$, to $9 \times 10^{13}$ at $D = 60$: the condition that selects the period weakens with $D$. The falling factors are those of a power law: with the harmonic set fixed ($D = 40$–$61$) the sensitivity is $n^{21.7}$ to 2%, and an exponential $e^{n/2}n^p$ misses by 17% (Sec. V explains the selection itself).

*Critical exponent.* For perturbations $\propto e^{\lambda\tau}$ every $\tau$-derivative in the linearized equations gains $+\lambda$; the growth-mode matrix $M(\lambda)$ of the discretized problem is singular at a growth rate, and $\gamma = 1/\lambda$. The double-double solver gives $M(\lambda)$ exactly (forward-mode duals), and $\gamma$ is the root of $\det M(\lambda)$ near its predicted value: 67 points for $8.25 \le D \le 61$, accurate to about $10^{-12}$ for $D = 16$–$58$ and agreeing with the double-precision values to $10^{-8}$–$10^{-12}$. The double-precision $\gamma$ scatters by up to $6 \times 10^{-5}$ above $D \approx 36$.

TABLE I. Checks of the double-double solver.

| Check | Result |
|---|---|
| double-double arithmetic against mpmath | $1.3$–$1.6 \times 10^{-32}$ relative |
| horizon series against 40-digit mpmath | $2 \times 10^{-32}$ |
| Newton at $D = 8$ | quadratic, $\lvert F\rvert = 5 \times 10^{-31}$ in 4 steps |
| $\Delta(7)$, $\Delta(7.75)$ against 39-mode double-precision references ($\pm 3 \times 10^{-11}$) | $+3.1 \times 10^{-12}$, $-2.8 \times 10^{-12}$ |
| 32 / 48 / 64 collocation points in $\tau$, $D = 7$–$17$ | $\Delta$ agrees to $10^{-16}$ |
| 32 / 48 collocation points, $D = 40$ | $9 \times 10^{-22}$ |
| harmonics pinned above 13 / above 15, $D = 40$, 50, 60 | $+1.5$, $-1.1$, $-1.2 \times 10^{-15}$ |
| second machine (separate build), $D = 59.84$–$60.09$ | $7.2 \times 10^{-18}$ rms |
| the 94 points above $D = 60.1$ against the $1/D$ fit on $D \le 60.1$ | $2.6 \times 10^{-15}$ |

## III. THE ECHO PERIOD AND CRITICAL EXPONENT FOR $D \ge 4$

### A. One formula for all $D \ge 4$

With $\varepsilon = 1/n = 1/(D-2)$,

$$\Delta(D) = \frac{\sum_{k=0}^{8} a_k \varepsilon^k}{1 + b_1\varepsilon + b_2\varepsilon^2}, \qquad \gamma(D) = \frac{\frac{1}{2} + \sum_{k=1}^{7} e_k \varepsilon^k}{1 + f_1\varepsilon + f_2\varepsilon^2}, \tag{3}$$

with the coefficients of Table II (11 fitted for $\Delta$ and 9 for $\gamma$), fitted on the double-precision points $4 \le D \le 30.2$. It is an empirical fit, not a law derived from the equations; no earlier formula for $\Delta(D)$ was found (Kol's analytic $2\pi/\Delta = \sqrt{(10-D)/(D-2)}$ is for the black-string merger, not for scalar collapse [20]), and the earlier fit for $\gamma$, $0.466\,(1 - e^{-0.408D})$ on $3.5 \le D \le 14$ [8], has a different form and limit. The formulas reproduce $\Delta$ to $5 \times 10^{-10}$ ($4 \le D \le 4.3$), $4.3 \times 10^{-9}$ (4.3–20) and $10^{-8}$ (20–30), and $\gamma$ to $3.4 \times 10^{-7}$, $3.3 \times 10^{-8}$ and $2.3 \times 10^{-9}$, each within the precision of the points. Against the double-double points, which the fit never saw, the formula holds to $2.7 \times 10^{-9}$ on 8.25–21, $7.5 \times 10^{-9}$ to $D = 30$ and $5.8 \times 10^{-8}$ to $D = 61$ (Fig. 2). Its constant $a_0$ is not $\Delta_\infty$: it lies $4 \times 10^{-6}$ above the double-double limit of Sec. IV, which its fitting range could not see. The limit $\gamma_\infty = 1/2$ is imposed in the $\gamma$ fit,

not derived (Sec. IV E tests it). The denominators vanish only at $D = -3.2, -11.0$ ($\Delta$) and $D = -13.3, -20.4$ ($\gamma$). Values are listed in Table III. Both $\Delta$ and $\gamma$ are monotonic up to $D = 61$: Sorkin and Oren's reading of their $D \le 11$ values as a minimum of $\Delta$ and a maximum of $\gamma$ near $D = 11$–$13$ [7] is not borne out.

TABLE II. Coefficients of Eq. (3).

| $k$ | $a_k$ | $b_k$ | $e_k$ | $f_k$ |
|---|---|---|---|---|
| 0 | $+1.6438975180$ | 1 | $+0.5$ (fixed) | 1 |
| 1 | $+4.0378045077 \times 10^{1}$ | $+1.8251453626 \times 10^{1}$ | $+1.8083629635 \times 10^{1}$ | $+3.7683686025 \times 10^{1}$ |
| 2 | $+2.5233285989 \times 10^{2}$ | $+6.8182050167 \times 10^{1}$ | $+1.5241588758 \times 10^{2}$ | $+3.4245505235 \times 10^{2}$ |
| 3 | $+2.0673502321 \times 10^{2}$ | | $-5.2341588541 \times 10^{1}$ | |
| 4 | $-3.5709850575 \times 10^{2}$ | | $+9.9259594505$ | |
| 5 | $+3.1347320996 \times 10^{2}$ | | $-9.8386450353 \times 10^{1}$ | |
| 6 | $-4.1891374574 \times 10^{2}$ | | $+7.4529506074 \times 10^{1}$ | |
| 7 | $+3.2695252782 \times 10^{2}$ | | $-4.8272358903 \times 10^{1}$ | |
| 8 | $-1.6050467699 \times 10^{2}$ | | | |

TABLE III. Values. $\Delta$ for $D = 10$–$60$ is double-double (interpolated to $10^{-14}$ where $D$ is not a computed point); $\Delta(8)$ is a separate double-double run; $\Delta$ for $D = 5$–$7$ and $\gamma$ for $D = 5$–$8$ are double-precision, $D = 4$ is double-double; $\gamma$ for $D = 10$–$60$ is double-double (interpolated where $D$ is not a computed point), and $\gamma(70)$, $\gamma(100)$, $\gamma(1000)$ extrapolate it with $\lambda_\infty = 2$ fixed. $\Delta(70)$ and $\Delta(100)$ are extrapolations on which all models fitting the double-double points agree (to $10^{-11}$, $2 \times 10^{-8}$); $D = 1000$ and $\infty$ assume analyticity in $1/D$.

| $D$ | $\Delta$ | $\gamma$ | $D$ | $\Delta$ | $\gamma$ |
|---|---|---|---|---|---|
| 4 | 3.445452402 | 0.37396072 | 20 | 2.112080154580 | 0.4737106 |
| 5 | 3.2217019 | 0.4131269 | 30 | 1.964957899258 | 0.48079268 |
| 6 | 3.0094832 | 0.4300795 | 40 | 1.888568197910 | 0.484776464 |
| 7 | 2.8427532 | 0.4399912 | 50 | 1.841659503007 | 0.487364941 |
| 8 | 2.711790054444 | 0.4467164 | 60 | 1.809889459975 | 0.4891910657 |
| 10 | 2.521078919562 | 0.4555879375 | 70 | 1.78693476494 | 0.490552 |
| 12 | 2.389188553641 | 0.4614055891 | 100 | 1.7450264844 | 0.493134 |
| 15 | 2.253098407194 | 0.4673655689 | 1000 | 1.65424086 | 0.499250 |
| | | | $\infty$ | $1.643893746 \pm 1.5 \times 10^{-8}$ | 1/2 (fits: $\pm 5 \times 10^{-7}$) |

### B. Comparison with published values

At $D = 4$ the formula gives $\Delta = 3.4454524$ and $\gamma = 0.3739610$, in agreement with $\Delta = 3.445452402(3)$ [3] to about $10^{-8}$ and with $\gamma = 0.373961 \pm 10^{-6}$ [12] to about $10^{-7}$. Below $D = 4$ polynomial fits to double-double points at $D = 3.70$–$3.80$ put the maximum of $\Delta$ at $D = 3.755725$, $\Delta = 3.46677169$ (spread over fits $6 \times 10^{-9}$), in agreement with $D = 3.755726$, $\Delta = 3.466772$ of [12] within their $\pm 10^{-6}$.

At $D = 5$, Ref. [12] gives $\Delta = 3.22176$ and $\gamma = 0.41322$, which are $5.8 \times 10^{-5}$ and $9.3 \times 10^{-5}$ above our values. Our $\Delta(5) = 3.2217019446$ comes from two independent codes that agree to $10^{-11}$: the double-precision solver at 31, 39 and 47 Fourier modes, extrapolated, and the double-double solver at 48 and 64 collocation points. The published data of [12] (repository CriticalSpacetimeCrystal) show their $D = 5$ value still moving by $-1.66 \times 10^{-4}$ between 256 and 512 modes, toward ours, in the range $D \gtrsim 4.8$ where they note that their light-cone expansion order must be increased; their offset from our values grows smoothly from $2.7 \times 10^{-7}$ at $D = 4$ (within their stated $\pm 10^{-6}$) to $5 \times 10^{-4}$ at $D = 5.5$, and at $D = 3.7$ the two agree to $8 \times 10^{-8}$, so there is no difference of convention. Their $\gamma$ list has no points for $4.79 < D < 5.49$, where they report that their perturbation method fails, so their $\gamma(5)$ is not a direct computation in the published data; ours, 0.41312694 (stable to $10^{-7}$ across resolutions), agrees with Bland and Kunstatter's $0.4131 \pm 0.0001$ [9]. Below $D = 4$ the double-double solver, with its matching nodes moved to $x = 0.005$ and 0.8, gives $\Delta(3.7) = 3.46520048 \pm 3 \times 10^{-8}$ and $\gamma(3.7) = 0.34865019 \pm 8 \times 10^{-8}$, converged in the $\tau$ resolution (48, 64 and 80 points); our 47-mode f64 points there are 2–5 $\times 10^{-6}$ high in $\Delta$. Extrapolated from $3.64 \le D \le 4$, $\gamma(3.5) = 0.3235 \pm 0.0030$, in agreement with [12] (0.32239) and not with $0.349 \pm 0.003$ in [8]. The modes needed grow from about 73 at $D = 4$ to about 110 at $D = 3.5$ and several hundred near $D = 3.05$; at $D = 3$ the scalar field decouples from gravity in these equations, so the family cannot reach a regular $D = 3$ limit.

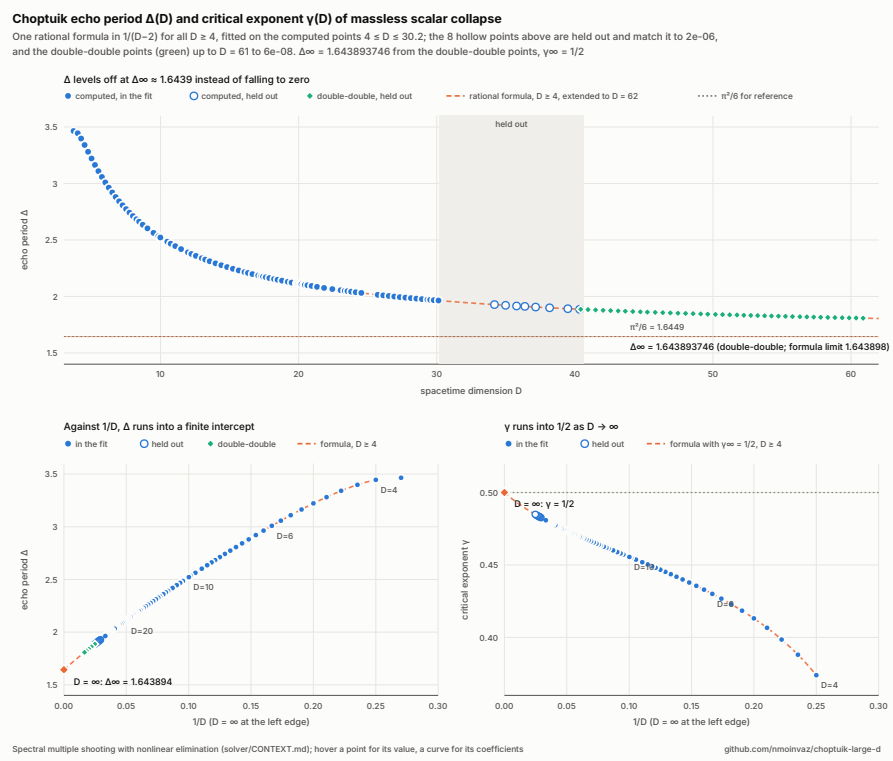



FIG. 2. $\Delta(D)$ and $\gamma(D)$ with the rational formula (3) (dashed), fitted on $4 \le D \le 30.2$; hollow points are held out, and green diamonds are double-double points up to $D = 61$.

### C. Physical reading

Each echo is a factor $e^{\Delta}$ smaller than the last: 31 at $D = 4$, 12 at $D = 10$, 5.2 at $D = \infty$. A near-critical collapse makes about $\gamma \ln(1/|p - p^*|)/\Delta$ echoes before it forms a black hole or disperses, so each further echo costs $\Delta/(\gamma \ln 10)$ decimal digits of fine-tuning: 4.0 at $D = 4$, 2.4 at $D = 10$ and 1.43 at $D = \infty$. In higher dimensions the echoes thus come closer together in scale and are cheaper to resolve. The shape of the echo simplifies too (Fig. 1): at $D = 4$ the centre slope is almost a square wave, with a third harmonic 44% of the first, while at $D = 60$ the third harmonic is 8% of the first, and the $D = \infty$ theory of Sec. V E gives 5.5%.

## IV. THE LIMIT $D \to \infty$

### A. Is the period analytic in $1/D$?

Each model in Table IV is a polynomial in one variable, fitted at 60 digits on the cleaned double-double points $10 \le D \le 45$ (or $\le 37.5$) and compared with every point above, up to $D = 60.1$, which it never saw (the data snapshot when the table and Fig. 3 were made). Every model fits its range to the level of the points, so the fit alone decides nothing; the unseen points do. Fitted on $D \le 45$, the $1/D$ series predicts $\Delta$ fifteen units of $D$ ahead to $6 \times 10^{-13}$, while the series in half-integer powers miss by 20 to 250 times more, and their limits jump with the degree. Adding one non-analytic term ($D^{-1/2}$, $D^{-3/2}$, $\ln D/D$, $\ln D/D^2$ or $e^{-D/4}$) to the degree-16 series neither degrades the prediction nor moves the limit outside its uncertainty, and the term's coefficient is not determined. The only structure the points show is a singularity on the negative $D$ axis, near $D \approx -9$ to $-11$.

### B. The limit and its error bar

Fitted on all cleaned points, power series in $1/D$ of degree 16–20 and in $1/(D-2)$ of degree 18–20 fit to the noise and give limits from 1.643893744 to 1.643893753 (snapshot $D \le 60.1$) and from 1.643893745 to 1.643893752 (all 823

TABLE IV. Held-out test of large-$D$ models (cleaned double-double points; snapshot $D \le 60.1$).

| Model | max residual, $10 \le D \le 45$ | worst error, $45 < D \le 60.1$ | fitted on $D \le 37.5$: worst error | limit |
|---|---|---|---|---|
| $1/D$, degree 16 | $4.1 \times 10^{-15}$ | $5.6 \times 10^{-13}$ | $4.2 \times 10^{-11}$ | 1.64389375 |
| $1/D$, degree 18 | $4.1 \times 10^{-15}$ | $1.6 \times 10^{-12}$ | $1.4 \times 10^{-10}$ | 1.64389377 |
| $1/(D-2)$, degree 18 | $4.0 \times 10^{-15}$ | $3.1 \times 10^{-12}$ | $7.7 \times 10^{-11}$ | 1.64389377 |
| $(D-2)^{-1/2}$, degree 20 | $4.0 \times 10^{-15}$ | $1.3 \times 10^{-11}$ | $6.6 \times 10^{-10}$ | 1.638 |
| $D^{-1/2}$, degree 20 | $3.9 \times 10^{-15}$ | $1.4 \times 10^{-10}$ | $5.3 \times 10^{-9}$ | 1.556 |

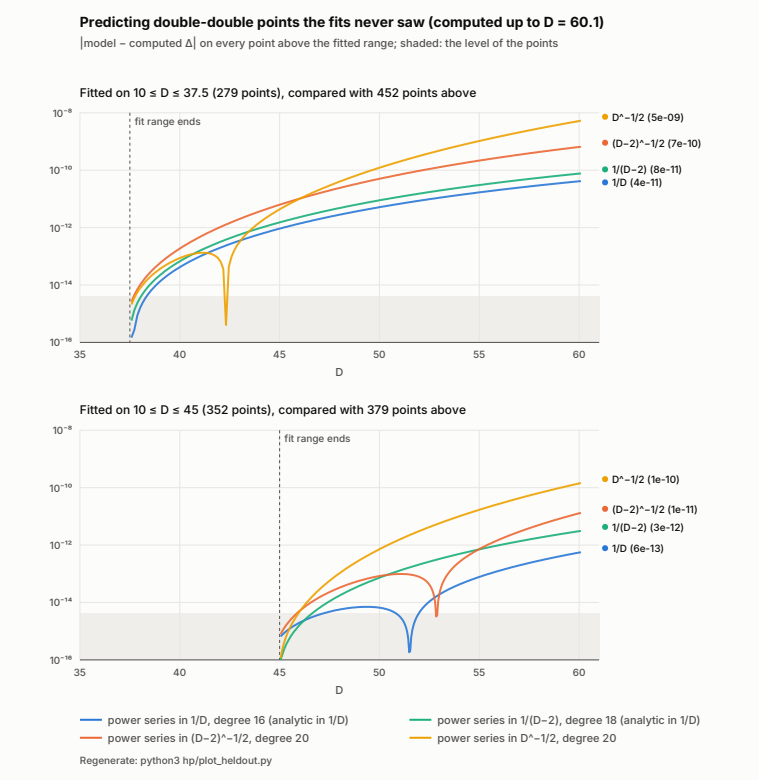



FIG. 3. |model − computed Δ| on every double-double point above the fitted range, for fits on $10 \le D \le 37.5$ (top) and $10 \le D \le 45$ (bottom); shaded: the level of the points.

cleaned points to $D = 61.03$). Over 48 choices of model ($1/D$, $1/(D-2)$), degree, fit range and cleaning, the limit ranges over 1.643893744–1.643893763, centred on 1.643893754; fitting on $D \le 45$ alone pins it only to $2.5 \times 10^{-7}$. A spread over fit choices is not an error with a stated coverage, so we tested the procedure on synthetic data with a known limit: the real $D$ grid, noise and junk points like the real ones, and added structures ranging from a distant pole to a complex pole pair and a logarithmic branch point near the data, plus one term outside the fitted model class. In all six cases the 48 fits put the known limit inside their range, with the centre 0.79–0.88 $\times 10^{-8}$ above it. A second test, whose planted truth is not made by the fitting code (the rational formula of Sec. III carrying the real data's residual pattern), finds $0.78 \times 10^{-8}$. We therefore lower the centre by $0.8 \times 10^{-8}$ and quote, if $\Delta$ is analytic in $1/D$,

$$\Delta_\infty = 1.643893746 \pm 1.5 \times 10^{-8}, \qquad \omega_\infty = 2\pi/\Delta_\infty = 3.82213590 \pm 4 \times 10^{-8}, \tag{4}$$

and $\Delta(D) \approx \Delta_\infty + 10.37/n + \dots$. The bar is an estimator range, not a calibrated error: it contains the planted limit in every graded case of both tests and the 48-fit range except its top $2 \times 10^{-9}$, which comes from an under-resolved fit. Other reasonable rules for choosing the fits, or removing the points the runs themselves flag, widen it to 0.7–1 $\times 10^{-7}$ (an independent review of the numerics, in the repository). Analyticity is the natural default, since the large-$D$ expansion of the equations has no other scale, but it is not proven: as Sec. V explains, the period is selected by an effect that the $1/D$ expansion does not see.

### C. The conjecture $\Delta \to 0$

The conjecture $\Delta \to 0$ [12, 16] comes with no rate. Fitted on $10 \le D \le 45$ and compared with $45 < D \le 61$: a power-law approach to 0, $D^{-p}$ times a series in $1/D$ with $p \ge 0.01$, misses the held-out points by $6 \times 10^{-8}$ or more, against $7 \times 10^{-13}$ for the analytic series; a series in $1/\ln D$ with zero limit misses them by $1.3 \times 10^{-11}$ and still gives $\Delta \approx 0.88$ at $D = 10^{30}$; adding a $c/\ln D$ term to the analytic series gives $c \approx 4 \times 10^{-7}$ and leaves the limit at 1.6438937. A power-law approach to 0 with $p \ge 0.01$ is excluded; a logarithmic one is disfavoured but cannot be excluded from $D \le 61$ alone, and neither can a crossover far beyond the data. Bland's hypothesis $\Delta_\infty = 5/2$ [10] is excluded by $\Delta(14) = 2.29$ and decreasing.

### D. Candidate closed forms

Five constants that a product search placed inside the window of the earlier double-precision estimate (about nine such coincidences are expected by chance in that search), and four classical constants, were tested. Each was forced as the limit of a power series in $1/D$ fitted on $10 \le D \le 45$ (degree 14 to 26) and compared with the 352 cleaned points from 45 to 59.8 (Table V). All nine are excluded, given analyticity in $1/D$: the closest sits $1.3 \times 10^{-5}$ from the limit, about 840 times its uncertainty, and every forced fit predicts the unseen points at least 100 times worse than the free fit. Nor is $\Delta_\infty$ a product of two of the listed constants with rational prefactor $i/j$ $(i, j \le 12)$ and exponents $\pm\frac{1}{2}, \pm 1, \pm 2$ (514 332 expressions; about 0.02 chance hits would be expected in the window, so this null has little power). At eight known digits no integer-relation search could support a closed form, so whether $\Delta_\infty$ has one is open.

TABLE V. Candidate closed forms for $\Delta_\infty$, forced as the limit: best held-out error and its ratio to the free fit. $G$ is Catalan's constant, $j_{\nu,k}$ the $k$-th zero of $J_\nu$.

| Candidate | value | minus $\Delta_\infty$ | held-out error (ratio) |
|---|---|---|---|
| none (free fit) | $1.643893746 \pm 1.5 \times 10^{-8}$ | | $4.8 \times 10^{-13}$ |
| $\frac{5}{7}\sqrt{G}\, j_{0,1}$ | 1.6439747 | $+8.1 \times 10^{-5}$ | $6.3 \times 10^{-11}$ (130) |
| $\frac{7}{5}(\ln 2)^{-2}\pi^{-1/2}$ | 1.6440014 | $+1.1 \times 10^{-4}$ | $1.1 \times 10^{-10}$ (230) |
| $\frac{7}{2}\sqrt{\ln 2/\pi}$ | 1.6440152 | $+1.2 \times 10^{-4}$ | $1.3 \times 10^{-10}$ (280) |
| $\frac{9}{8}\sqrt{j_{2,1}/j_{0,1}}$ | 1.6440207 | $+1.3 \times 10^{-4}$ | $1.4 \times 10^{-10}$ (300) |
| $\frac{10}{9}\sqrt{j_{0,1}/\ln 3}$ | 1.6439063 | $+1.3 \times 10^{-5}$ | $5.3 \times 10^{-11}$ (110) |
| $\pi^2/6$ | 1.6449341 | $+1.0 \times 10^{-3}$ | $1.7 \times 10^{-9}$ (3500) |
| $2\pi/j_{1,1}$ | 1.6397880 | $-4.1 \times 10^{-3}$ | $7.0 \times 10^{-9}$ (15000) |
| $\sqrt{e}$ | 1.6487213 | $+4.8 \times 10^{-3}$ | $8.1 \times 10^{-9}$ (17000) |
| $(1+\sqrt{5})/2$ | 1.6180340 | $-2.6 \times 10^{-2}$ | $4.4 \times 10^{-8}$ (92000) |

### E. The critical exponent tends to 1/2

Unconstrained fits of the double-precision $\gamma$ on $D \ge 8$–12 give $\gamma_\infty = 0.4996$–$0.4998$. The double-double $\gamma$ pins the limit much more tightly. In free fits of $\lambda = 1/\gamma$ in powers of $1/n$, $\lambda_\infty - 2$ falls by a factor of about 3.5 with each added power: on $D \ge 30$, $2.4 \times 10^{-4}$, $6.7 \times 10^{-5}$, $1.9 \times 10^{-5}$, $5.5 \times 10^{-6}$ and $(1.6 \pm 2.2) \times 10^{-6}$ for four to eight powers, with no sign of levelling off (Fig. 4), and at every order fixing $\lambda_\infty = 2$ fits better than leaving it free with the same number of parameters. So $\lambda_\infty = 2$ to about $10^{-6}$ and $\gamma_\infty = 1/2$ to about $5 \times 10^{-7}$, with

$$\lambda = 2 + \frac{3.0324}{n} - \frac{34.3}{n^2} + \dots, \qquad \gamma = \frac{1}{2} - \frac{0.7581}{n} + \dots \tag{5}$$

(the $1/n$ coefficient is stable to about $10^{-3}$ across fit choices: 3.0313–3.0324). At $D = 61$, $\gamma = 0.48935$, and the points keep rising, which excludes the limit 0.466 of the fit of [8]. The value 1/2 itself is not new: it is Rozali and Way's estimate [14], and is discussed in [12, 15, 16]. Like $\Delta_\infty$, it is not fixed at any finite order of the $1/n$ expansion (Sec. V B); its explanation is in Sec. V D.

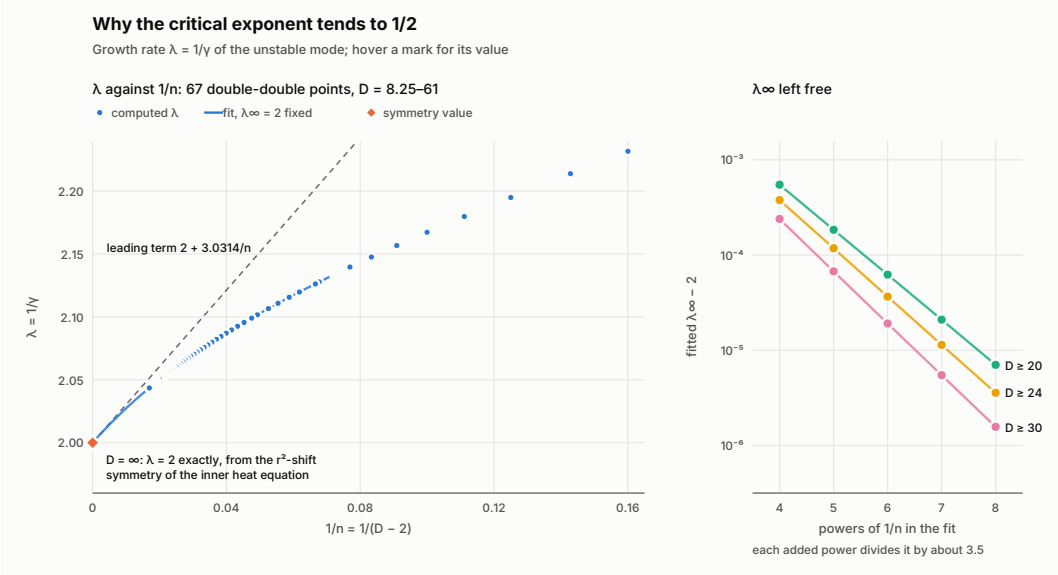



FIG. 4. Left: $\lambda = 1/\gamma$ against $1/(D-2)$ for 67 double-double points, with a fit fixing $\lambda_\infty = 2$ (solid, over its range $D \geq 16$) and its leading term $2 + 3.03/n$ (dashed); at $D = \infty$, $\lambda = 2$ is an exact growth rate of the inner problem, from the $r^2$-shift symmetry of the inner heat equation (Sec. V D). Right: with $\lambda_\infty$ left free, the fitted $\lambda_\infty - 2$ falls about 3.5-fold per added power of $1/n$, for three fit ranges.

## V. THE LARGE-$D$ PICTURE

The numerics fix $\Delta_\infty$ and $\gamma_\infty$; this section asks where they come from in the equations. The theory is developed, with proofs where they exist, in the companion paper [18]; here we state its results as physics and give the numerical evidence for each.

### A. How the solution scales with $D$

On the double-double solutions at $D = 12$–$60$: $p \approx xF(\theta)/\sqrt{n}$ near the centre, and $p + m$ is smaller by a further factor of order $1/n$; $c = 1 - O(1/n)$ and $b - 1 = (p^2 + m^2)/(n-1)$ to about 3%; near the light cone $c$ stays within $10^{-3}$ of 1. The profile $F = \sqrt{n}\,\partial_x p(x=0)$ converges as $D$ grows (Table VI). So the field and its gravity weaken inside the light cone: the centre slope falls like $2.72/\sqrt{n}$ and the departure of $c$ from 1 at the centre like $3.65/n$, and the last two columns of the table approach the same limit, as the large-$D$ expansion requires $(c = 1 - \varepsilon F^2(1-x^2) + \dots)$. At $D = \infty$ the echo inside the light cone is therefore a free scalar wave on flat space. (In four dimensions Price and Pullin approximated the critical solution, in each of two regions, by flat-space scalar-field solutions [23].)

TABLE VI. The echo profile $F = \sqrt{n}\,\partial_x p(0)$ and the strength of gravity at the centre on the double-double solutions, and their $D = \infty$ extrapolations.

| $D$ | first harmonic of $F$ | $\lvert F_3/F_1\rvert$ | $\lvert F_5/F_1\rvert$ | $n(1 - \langle c(0)\rangle)$ | $\langle F^2\rangle$ |
|---|---|---|---|---|---|
| 20 | 3.480 | 0.130 | 0.024 | 3.021 | 6.162 |
| 30 | 3.212 | 0.105 | 0.016 | 3.201 | 5.218 |
| 40 | 3.086 | 0.093 | 0.012 | 3.304 | 4.804 |
| 50 | 3.011 | 0.085 | 0.0097 | 3.372 | 4.568 |
| 60 | 2.962 | 0.080 | 0.0085 | 3.421 | 4.416 |
| $\infty$ (extrapolated) | $\approx 2.72$ | 0.0549–0.0552 | 0.0032–0.0034 | $\approx 3.65$ | $\approx 3.7$ |

### B. A free wave cannot echo, and the period is selected beyond all orders

At leading order ($c = b = 1$) a single harmonic $e^{i\mu\theta}$ of the field, with $s = i\mu\omega$, solves Gauss's hypergeometric equation: the solution regular at the centre is $p-m = x\,{}_2F_1\big(\frac{1+s}{2}, \frac{2+s}{2}; \frac{n+1}{2}; x^2\big)$, and by Gauss's connection formula [21] it contains, near the light cone, the non-analytic branch $(1-x^2)^{n/2-1-s}$ with coefficient

$$C(n,s) = \frac{\Gamma\big(\frac{n+1}{2}\big)\,\Gamma\big(s+1-\frac{n}{2}\big)}{\Gamma\big(\frac{1+s}{2}\big)\,\Gamma\big(\frac{2+s}{2}\big)}, \tag{6}$$

which is finite and non-zero for every real $\omega \neq 0$, odd $\mu$ and $n \geq 2$ [18, Prop. 2]. So a free wave that is periodic in $\tau$ and regular at the centre is never analytic at the light cone: on its own it cannot echo, in any $D \geq 4$. (The free wave equation in similarity coordinates was analysed in 3+1 dimensions by Donninger [22], which contains the $D = 4$ case.) For the fundamental at $\omega_\infty$, $|C| = 0.0114$–$0.0405$ for $n = 18$–$1000$ [18]: small, through a factor of order $e^{-\pi\omega/2}$, but never zero. Gravity has to cancel this coefficient harmonic by harmonic, and that cancellation is what selects $\Delta$ and the profile $F$.

The $1/n$ expansion of the field equations never sees it. Within the scaling measured above (integer powers of $1/n$ after an overall $n^{-1/2}$ in the field), the expansion can be carried to all orders for *any* periodic profile $F$ and *any* period, regular at the centre and analytic at the light cone at every order [18, Prop. 1]; the leading-order part of this freedom, any periodic profile with any period, is in Ecker, Ecker and Grumiller [16]. Numerically: substituting the expansion through third order into the exact equations, with $F$ from the $D = 60$ solution, the relative residuals fall like $n^{-3}$ for $n = 50$–$800$, both for $\omega_0 = 2\pi/1.64389$ and for a wrong $\omega_0 = 2\pi/1.4$. The same holds for growing perturbations $e^{\lambda\tau}$ with any $\lambda$. So if $\Delta(D)$ and $\gamma(D)$ are analytic in $1/D$, their expansion coefficients cannot be computed order by order from the field equations: they are selected by effects smaller than every power of $1/n$. The solver does not see them as exponentially small: it matches at $x = 0.7$, where the fast mode $e^{-n\Phi(x)}$, $\Phi = \ln x - \frac{1}{2}\ln(1-x^2)$, is of order one ($|e^{-n\Phi}| = 1$ on $\operatorname{Re} x^2 = \frac{1}{2}$), and its sensitivity grows like a power of $n$ (Sec. II). Non-perturbative, trans-series corrections in $1/D$ are familiar in large-$D$ gravity [13]; here they decide the period itself.

The non-analytic piece is a fast WKB mode, $e^{-n\Phi(x)}$ with $\Phi = \ln x - \frac{1}{2}\ln(1-x^2)$, which is exponentially small near the light cone ($x > 1/\sqrt{2}$) and is the singular branch $x^{2-n}$ at the centre. Gravity acts on it at order one [18, Prop. 3]: along its characteristics it multiplies the mode by a factor that never vanishes (about 40 over $0.72 \leq x \leq 0.9$ at the $D = \infty$ amplitude, checked by direct integration with errors falling like $1/n$) and shifts its light-cone exponent by $\langle F^2\rangle \approx 3.7$, so an expansion in the wave amplitude cannot capture the selection, and gravity cannot cancel the piece on its way to the light cone. Every characteristic of the mode reaches the centre, where it is singular, so a solution regular at the centre carries none of it inside the light cone, at any order of the WKB expansion. The late orders of the leading-order (gravity-free) $1/n$ series locate its origin [24], [18, Prop. 4]: $C(n,s)$ is the Stokes multiplier of a mode born at infinite self-similar radius and switched on across a Stokes line along $x > 1$ that ends at the light cone. The period is therefore decided outside the past light cone, where the fast mode and the slow solution meet: at $x \sim \sqrt{n}$.

### C. The inner region: Rozali and Way's heat equation

At $x \sim \sqrt{n}$ the expansion is not uniform: $\varepsilon x^2 F^2 = O(1)$, so gravity is of order one there, and fast and slow modes are not separated. In physical variables this is the region $r = O(\sqrt{n})$ at fixed $t$, which is Rozali and Way's regime [14] (they keep $r$ fixed and rescale time by $\sqrt{D}$). Put $\rho = r/\sqrt{n}$,

$$s = \frac{\rho^2}{2} = \frac{r^2}{2n}, \qquad p - m = \sqrt{s}\,Y, \qquad \sqrt{n}\,(p+m) = \sqrt{2}\,s\,W, \tag{7}$$

with $t$ the time and $c$ the lapse-type function as before (this $s$ is unrelated to the $s = i\mu\omega$ in $C(n,s)$, Sec. V B). At leading order in $1/n$ the field equations become [18, Prop. 5(i)]

$$Y_t = c\,W, \qquad W_t = c\Big(Y_s + \frac{Y^3}{4}\Big), \qquad c_s = \frac{c\,Y^2}{4}, \qquad \text{i.e.} \qquad Y_s = \frac{1}{c}\,\partial_t\Big(\frac{1}{c}\,\partial_t Y\Big) - \frac{Y^3}{4}, \tag{8}$$

with $b - 1 = (p-m)^2/(2n)$: a nonlinear heat equation in which $s = r^2/(2n)$ plays the role of time and $t$ that of space, with no large parameter left. These are the leading-order large-$D$ equations of Rozali and Way, who noted this parabolic character, in other variables (the map between the two, checked symbolically, is in the proof of Prop. 5(i) of [18]). A discretely self-similar solution depends on $\xi = \rho/(-t) = x/\sqrt{n}$ and $\theta$; its series at $\xi = 0$ reproduces the

top-degree terms of the outer $1/n$ expansion (to $2\times10^{-13}$ at orders 1–10). The outer fast mode $e^{-n/(2x^2)}=e^{-t^2/(4s)}$ is the heat kernel, and without gravity the inner problem is Weber's equation, whose classical Stokes constant [21] reproduces $C(n,s)$ at large $n$ [18]: the light-cone obstruction is a property of this $n$-free problem. What this work adds to Rozali and Way's equations is the discretely self-similar problem posed on them: the selection rule (Sec. V E) and the $r^2$-shift argument for $\gamma_\infty$.

### D. The $r^2$-shift and $\gamma_\infty = 1/2$

Equations (8) contain no explicit $s$ or $t$. They are invariant under $t$-translation, $s$-translation, the scaling $(t,s,Y,W)\to(at,a^2s,Y/a,W/a^2)$, and the time reparametrizations $t\to T(t)$, $c\to T'c$, which are gauge. A self-similar solution is invariant under (a discrete subgroup of) the scaling, and a continuous symmetry of scaling weight $k$ acting on it produces a linear perturbation $\propto e^{k\tau}$, because $\partial_t\propto e^{\tau}$ and $\partial_s\propto e^{2\tau}$ at fixed $\xi$, $\theta$. The $t$-translation ($k=1$) is the familiar $\lambda=1$ mode, a shift of the time of the singularity. The $s$-translation ($k=2$) gives $\lambda=2$. It moves the lapse at the centre, which the normalization $c=1$ there forbids, but composed with the time reparametrization $t\to t-e^{\tau}P(\theta)$, where $P+\omega P'=-2F^2$, it leaves the centre lapse alone. The combination is an exact perturbation of the $D=\infty$ inner solution with $\lambda=2$, regular at the centre [18, Prop. 5(ii)]. As a check, a perturbation recursion with gravity that builds in nothing about symmetries, run at $\lambda=2$, reproduces the mode to $10^{-180}$ in every resolved harmonic, while $\lambda=1.9$ or 2.1 miss by 7.5% at the third order. Because the mode is the derivative of a family of exact solutions (shifted and reparametrized copies of the background), its fast-mode Stokes multipliers vanish whenever the background's do, wherever these multipliers are defined (by Borel summation, which is not yet proved in general) [18, Prop. 5(iii)]: if the selection rule of Sec. V E picks the background, $\lambda=2$ is one of its growth rates, with no transcendental condition involved.

The mode is physical, not gauge: the reparametrization is pure gauge and $t$-translation only shifts the accumulation time, but $s$-translation shifts $r^2$, with $r$ the areal radius. In the inner region critical collapse is diffusion in $s=r^2/(2n)$, and the unstable mode moves the solution along $s$. A supercritical evolution therefore forms its horizon at $r_{\rm BH}^2\propto p-p^*$, so $r_{\rm BH}\propto(p-p^*)^{1/2}$ and $\gamma_\infty=1/2$: the discretely self-similar counterpart of Rozali and Way's estimate [14], which came from the divergence radius of their oscillating solutions. This also explains why $\lambda_\infty$ is a simple number while $\omega_\infty$ is not: $\omega$ is fixed by the vanishing of a Stokes multiplier, $\lambda$ by a symmetry. The statement is conditional [18, Cor. 1]: *if the $D=\infty$ critical solution is selected by the rule of Sec. V E, has no unstable mode other than $\lambda=2$ and the gauge mode $\lambda=1$, and $\lambda(D)$ is continuous at $1/n=0$, then $\gamma_\infty=1/2$.* The uniqueness of the unstable mode at $D=\infty$ is not shown. At finite $D$ the outer region and the $1/n$ corrections to Eqs. (8) depend on $r$ explicitly, so the symmetry is broken and the mode survives as an ordinary unstable mode with $\lambda=2+O(1/n)$; the data give $2+3.0324/n$ [Eq. (5)], a coefficient that would need the $O(1/n)$ inner equations.

### E. The selection rule at $D=\infty$

The $D=\infty$ form of the condition that the solution be analytic at the light cone is conjectured to be [18, Conj. 1]: *the $D=\infty$ critical solution is the discretely self-similar solution of the inner problem (8), regular at the centre, whose fast-mode Stokes multipliers vanish.* That is one complex condition per odd harmonic, for unknowns that are the harmonics of the profile $F$ (less one phase) and $\omega$, so the count balances; it uses no input from finite $D$. Without gravity it reduces to $C=0$, which has no solution, so the selection is gravity's doing even at $D=\infty$. The companion paper states the hypotheses under which the rule should follow from analyticity at the light cone, the points where a proof is still missing (its Conjecture 1 and Theorem B), and the numerical evidence for the rule [18], which we summarize.

Evaluated on the $D=\infty$ equations alone, with every harmonic kept exactly order by order in the amplitude, the rule gives the shape of the echo and its period. With the higher-harmonic multipliers tuned to zero at $A=2.72$ and $\omega=\omega_\infty$, the profile has $|F_3/F_1|=0.0551$ and $|F_5/F_1|=0.00314$, against 0.0549–0.0552 and 0.0032–0.0034 extrapolated from the solutions (Table VI); and the harmonic-1 multiplier has a real zero at amplitude $A_0=2.720$ at twelve orders (the solutions extrapolate to 2.70–2.72) and, at thirteen orders, at $\omega/\omega_\infty=0.99992$–$0.99998$, that is

$$\Delta_\infty = 1.6440\pm0.0001 \tag{9}$$

from the $D=\infty$ inner equations alone, against 1.643893746 from the finite-$D$ solutions. The $\pm0.0001$ is the range of estimators calibrated on planted functions of the same class, not an error bound, and it assumes the selection function has no other zeros near the selected one. The thirteenth coefficient rejected every alternative with such zeros constructed so far, but cannot exclude a larger ring [18]. The finite-$D$ value lies $4\times10^{-5}$ below the range and inside

the wider single-estimator range 1.6438–1.6442. The leading-order ratio of successive tuned harmonics, $|q|A^2 \approx 0.063$, is the same for every harmonic, and summed in the amplitude the tuned profile decays by about 0.057 per pair of harmonics, a rate of 1.43 per harmonic, within the predicted $D = \infty$ band of Fig. 5. The $D = \infty$ echo profile is then not an entire function of $\theta$: it has singularities at a finite distance, $|\mathrm{Im}\,\theta| \approx 1.4$, from the real axis.

Truncated to $M = 1$, 3 and 5 harmonics, the rule selects frequencies that approach $\omega_\infty$ with alternating offsets [18]. For the one-harmonic truncation, whose zero is not the physical one, the companion paper certifies (computer-assisted, following the template of Baldomá et al. [25], conditional on Braaksma's summability theorem [26]) that the selection condition has exactly one zero in a small box around it, and that it is simple [18, Thm. A] (Reiterer and Trubowitz gave a computer-assisted existence proof of the $D = 4$ critical solution itself [27]).

The matter models of Sec. VI give two further, truncation-free tests of the rule.

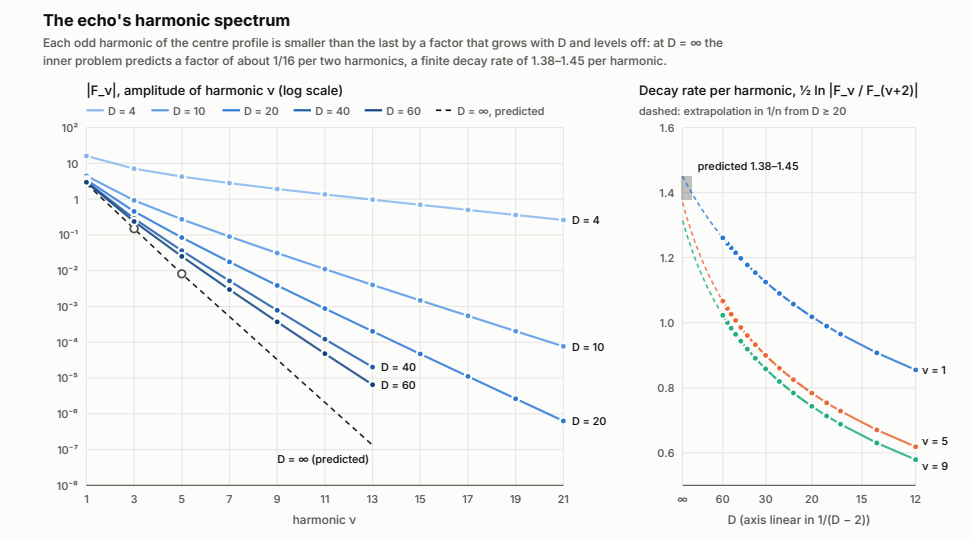



FIG. 5. Left: amplitude $|F_\nu|$ of each odd harmonic of the echo profile for $D = 4$–60 and the $D = \infty$ prediction of the inner problem (dashed; beyond $\nu = 5$ the leading-order ratio, conjectured). Right: the decay rate per harmonic, $\frac{1}{2}\ln|F_\nu/F_{\nu+2}|$, against $1/(D-2)$, with extrapolations from $D \geq 20$ and the predicted $D = \infty$ band.

### F. The next-to-leading-order condition of Ref. [16]

Ecker, Ecker and Grumiller use the same gauge: their $f = \alpha/a$ is our $c$, with $f = 1$ at the self-similarity horizon $x = 1$, and their profile is $\beta(\tau) \approx \sqrt{2/n}\,F(\omega\tau)$. At next-to-leading order they require $f \leq 1$ near the horizon at the zeros of $\beta$, which leads to their Eq. (20); in our variables the condition is $\omega^2 F'' + 3\omega F' = 0$ at the zeros of $F$. This condition does not select $\Delta$. The inequality $f \leq 1$ holds for every regular solution, whatever the period: from $xb_x = b(p^2 + m^2) - (n-1)b(b-1)$ with $b(0) = 1$, $b$ can never drop below 1, so $xc_x = (n-1)c(b-1) \geq 0$ and $c \leq c(1) = 1$. In the large-$D$ expansion, near a zero of $F$,

$$b - 1 = 2\varepsilon^2 x^2 F(\theta - \delta\theta)^2 + 2\varepsilon^4\omega^2 x^4 F'^2 + \ldots, \qquad \delta\theta(x) = -\varepsilon\omega x^2\,\frac{3F' + \omega F''}{2F'}, \tag{10}$$

so the next-to-leading-order term is the cross term of a square whose square term arrives one order later. The double-double solutions bear this out: they violate the condition $[(\omega^2 F'' + 3\omega F')/(3\omega F') = 0.98$–$1.06$ at $D = 12$–60, since $F$ is nearly antisymmetric about its zeros], the zero line of $p - m$ tilts as $\delta\theta(x)$ predicts (measured over predicted at $x = 1$: 0.69 to 0.88 for $D = 20$–60), and the maximum of $c$ at the centre approaches $1 - (\varepsilon^3/2)\omega^2 F'^2$, their Eq. (22) in this scaling (ratio 0.36 to 0.86). The same condition is used by Lai and Wen [28], who take it to select the period; the argument applies to that use as well.

## VI. OTHER MATTER AT LARGE $D$

Does the mechanism behind $\gamma_\infty = 1/2$ carry over to other matter? The deciding term is a centrifugal or potential term $V/r^2$ in the field equation: in the inner variable $s = r^2/(2n)$ it becomes $V/(2ns)$, so a $V$ of order $n$ leaves an

order-one $1/s$ term in the $D = \infty$ inner equations, which breaks the $s$-translation. The inner problems of several models were derived and checked symbolically (Table VII). For the multiplet with $\ell \geq 1$ and for wave maps, a search for Lie point symmetries of the inner equations (with polynomial infinitesimals of bounded degree) finds no replacement for the $s$-translation: the only exact positive growth rate left is the time translation, $\lambda = 1$. For these models $\gamma_\infty$ is not fixed by a symmetry, and $\gamma_\infty \neq 1/2$ is the generic expectation (a conjecture). The predictions were tested numerically as follows; each solver reproduces the published four-dimensional values first.

TABLE VII. The $D = \infty$ inner problem of several matter models and what it implies for $\gamma_\infty$.

| Model | $D = \infty$ inner problem | $s$-shift | expectation for $\gamma_\infty$ |
|---|---|---|---|
| real scalar | autonomous heat system (8) | yes | 1/2 (data: 1/2 to $5 \times 10^{-7}$) |
| complex scalar, axion–dilaton | the same, plus a target-curvature term | yes | 1/2 (axion–dilaton to $D = 80$: 1/2 to about $3 \times 10^{-6}$) |
| scalar multiplet, $\ell \geq 1$ | heat system plus $\ell\varphi/(2s)$ | broken | not fixed by a symmetry ($\ell = 1$ to $D = 9$: undecided) |
| Yang–Mills (type II) | that of the $\ell = 1$ multiplet | broken | the $\ell = 1$ value |
| wave maps ($\sigma$-model), coupling $\eta$ | a $\sin 2u/(4s)$ term | broken | $\eta$-dependent (CSS branch ends near $D = 8.5$) |

### A. Axion–dilaton and complex scalar

The critical solutions of the Hirschmann–Eardley family [29], whose $\kappa = 1$ member is the axion–dilaton system [30] and $\kappa = 0$ the free complex scalar [31], are spherically symmetric scalars with no potential. Their critical solutions rotate in phase and are continuously self-similar, $\phi = (-t)^{i\omega} f(x)$ for the complex field $\phi$, so they solve ordinary differential equations, with $\omega$ selected by analyticity at the past light cone. At $D = 4$ our codes reproduce $\omega = 1.176952724$ and $\gamma = 0.2641066$ for the axion–dilaton [30], and $\omega = 1.9154447$ and $\gamma = 0.3871055$ for the free complex scalar [31, 32]. The axion–dilaton solution was followed to $D = 80$ by complex-path shooting (with collocation as an independent check to $D \approx 8$), with its growth rate at each $D$. Its $\gamma$ rises monotonically from 0.2641 at $D = 4$ to 0.4896 at $D = 80$. In free fits of $\lambda$ in powers of $1/n$, $\lambda_\infty - 2$ falls about threefold per added power, to $8.6 \times 10^{-6}$, and at every order fixing $\lambda_\infty = 2$ fits better: $\lambda = 2 + 3.60/n - 27/n^2 + \ldots$, i.e. $\gamma = 1/2 - 0.900/n + \ldots$ (against $1/2 - 0.758/n$ for the real scalar), so $\gamma_\infty = 1/2$ to about $3 \times 10^{-6}$, an extrapolation from $D \leq 80$. The solution has the real scalar's large-$D$ structure: gravity is weak inside the light cone, the field is of order one and varies on $x \sim \sqrt{n}$, and there the equations become the heat system (8) for a complex field, with the target curvature as one extra term and no explicit $s$, so the $r^2$-shift gives $\lambda = 2$; the finite-$D$ solutions continued past the light cone converge to the $D = \infty$ inner solution with $O(1/n)$ corrections. Unlike for the real scalar, the uniqueness of the unstable mode was checked at finite $D$: besides the exact symmetry modes there is a single growing mode, from the full collocation spectra at $D = 4$–8 and from winding numbers of the growth-rate determinant at $D = 4$, 10, 20 and 40 (in boxes up to $\mathrm{Re}\,\lambda = 4.5$–6, $|\mathrm{Im}\,\lambda| = 1.2$–2.9). Finally, the selection rule passes a test without harmonic truncation, because the phase-rotating solution closes on one harmonic: the zero of the $D = \infty$ Stokes jump gives $\omega_\infty = 3.1998939$ for the axion–dilaton and 3.9789695 for the free complex scalar, and the finite-$D$ frequencies extrapolate to 3.1994 and 3.9784: before rounding, an agreement to $1.6 \times 10^{-4}$ and $1.3 \times 10^{-4}$ (relative).

Two caveats. The published axion–dilaton values for $5 \leq D \leq 10$ [33] are not reproduced: shooting with those parameters reaches the light cone, but not analytically, so they appear to satisfy a weaker horizon condition; we have not checked this against the authors' code. And at $D = 4$ the free complex scalar has a second, oscillatory unstable mode [32], which was not followed in $D$, so for $\kappa = 0$ it is open whether the solution is critical at large $D$. The large-$D$ CSS solution of Clark and Pimentel [15], whose radius exponents tend to 0, is a different solution (a real scalar with no phase rotation, no regular centre and no analytic light cone), so it does not contradict the mechanism; it shows that the mechanism needs a regular centre, an analytic light cone and the inner scaling, as well as the absence of a $1/r^2$ term.

### B. The $\sigma$-model: the CSS branch ends near $D = 8.5$

For the self-gravitating hedgehog wave map into $S^{D-1}$, with the dimensionless coupling $\eta = 4\pi G f^2$, the CSS solver reproduces the published four-dimensional values (e.g. $\gamma = 0.179$–$0.188$ for $\eta = 0.10$–$0.14$ against $\approx 0.18$ [34]; also [35–37]). At fixed $\eta = 0.14$ the CSS critical solution has one growing mode all along its branch, and $\gamma$ rises monotonically from 0.188 at $D = 4$ to 0.5135 at a fold at $D = 8.5444$, where the branch ends. Without gravity ($\eta = 0$) the branch

instead collapses onto the singular equator cone as $D \to 8$, with $\gamma \to 1/2$; that value is inherited from the cone, not from an $r^2$-shift. So this branch never reaches large $D$ and cannot test $\gamma_\infty$; the test needs the $\sigma$-model's DSS branch, which has not been computed. A by-product checks the selection rule in a second model: at $\eta = 0.14$ the zero of the $D = \infty$ Stokes multiplier for the CSS ground state, $A_0 = 1.9225686$, matches the finite-$D$ ground state extrapolated to $D = \infty$, $1.9226 \pm 0.0003$.

### C. The $\ell = 1$ scalar multiplet: undecided

A multiplet of massless scalars $\phi_m = \psi(t, r)\, Y_m$, with $Y_m$ the degree-$\ell$ harmonics on $S^n$, has a spherically symmetric stress tensor and a centrifugal term $\ell(\ell + n - 1)\psi/r^2$ of order $n$. The DSS solver, adapted to it and checked against the main solver at $\ell = 0$ (to rounding, $9 \times 10^{-14}$), reproduces the published $D = 4$ values for $\ell = 1$: $\Delta = 0.4598611$ and $\gamma = 0.118900$, against $0.460 \pm 0.002$ and $0.119 \pm 0.001$ [38]. Followed up in $D$:

| $D$ | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|
| $\Delta$ | 0.4598611 | 0.7833439 | 0.9115281 | 0.9639878 | 0.9837905 | 0.988205 |
| $\gamma$ | 0.118900 | 0.215509 | 0.265565 | 0.295733 | 0.316203 | 0.331286 |

Beyond $D = 9$ the period is selected ever more weakly ($|\partial\Delta/\partial F| \approx 4 \times 10^5$, a level the $\ell = 0$ solution reaches only near $D = 21$), and the continuation in double precision stops there. Six points do not decide $\gamma_\infty$: free fits put $\lambda_\infty$ anywhere from 0.55 to 3.95, and fixing $\lambda_\infty = 2$ fits as well. Compared with $\ell = 0$ at the same $D$, $\lambda_1 - \lambda_0$ falls roughly like $5.6/n$, and windowed fits of its limit bracket $\lambda_{1,\infty} - 2$ heuristically in 0.09–0.31 ($\gamma_\infty \approx 0.43$–$0.48$). That leans toward the expected $\gamma_\infty \neq 1/2$, but it is not a result: settling it needs $\ell = 1$ out to $D \approx 15$–$20$. The period peaks near 0.988 at $D \approx 9$ and then falls, so on present evidence its limit is finite and below the $\ell = 0$ value.

## VII. DISCUSSION

The numerics favour a finite $\Delta_\infty$ strongly, under the assumption of analyticity in $1/D$ that they support but cannot prove, and give $\gamma_\infty = 1/2$. One assumption and the open problems remain.

*One unstable mode.* At $D = 30$, 45 and 61 an argument-principle count of the zeros of the growth-mode determinant in $-0.08 < \mathrm{Re}\,\lambda < 6.18$, over a full period in $\mathrm{Im}\,\lambda$ (both symmetry sectors), finds besides $\lambda_{\mathrm{phys}}$ only the $\tau$ translation ($\lambda = 0$) and a gauge mode of the formulation at $\lambda = 1 + \langle c_1 \rangle$; a planted-zero test of the same code counts correctly in all 8 cases. A second resolution shows one extra pair at $D = 30$, absent at the production resolution and with the high-harmonic cut, which we take as a truncation artifact pending a third resolution. Lower $D$ with complex $\lambda$, larger $\mathrm{Re}\,\lambda$ and non-spherical perturbations are not covered, and no collapse has been evolved above $D = 14$, so a continuously self-similar solution [15] taking over the threshold is not excluded; in that case $1/\lambda$ would not be the critical exponent. For the axion–dilaton the corresponding check was made at $D \leq 40$ (Sec. VI).

*Open.* Higher orders of the selection rule, which would give $\Delta_\infty$ (and possibly a closed form) to more digits; a proof of the rule (Conjecture 1 of [18]); the uniqueness of the $\lambda = 2$ mode at $D = \infty$ and its $1/n$ coefficient; an independent check of the double-double points above $D = 40$; $\gamma$ just below $D = 4$; and the $\ell = 1$ multiplet beyond $D = 9$ and the $\sigma$-model's DSS branch, which would show whether a centrifugal term moves $\gamma_\infty$ away from 1/2.

## VIII. HOW THIS WORK WAS DONE

N. M. chose the problem, set the direction at each step and ran the long computations on his own machine. The code, the computations, the derivations, the proofs and the text, including this paper, were produced by Claude, Anthropic's AI model, working as a set of coordinated agents. N. M. has not checked the mathematics himself, and no human expert has reviewed the work.

*Checks of the results.* Two independent solvers agree where they overlap; two machines with separate builds agree to $7 \times 10^{-18}$ rms; an independent review re-derived the solver's equations and re-evaluated its residual in 200-bit arithmetic; every fit table reproduces from the data with the stated command; synthetic data with known limits test the error bar on $\Delta_\infty$; the large-$D$ expansion and the inner equations were re-derived by separate agents (with sympy where possible); the inner Stokes multiplier was computed by two implementations that share no code; and values at $D = 4$ and the maximum of $\Delta$ agree with the literature, and each matter solver reproduces the published four-dimensional values before it is continued in $D$.

*Checks of the text.* A separate agent wrote a simulated referee report on the combined draft from which this paper and its companion [18] were split. The citations of that draft were checked against publisher or arXiv records, and its novelty claims against the closest prior work, read rather than recalled; the references added since were checked by web search against arXiv, publisher or catalogue records, and the fields that could not be confirmed are marked. After the split, the numbers in the two papers were cross-checked against the repository (every number of three or more significant digits was found in a repository file), and the two papers against each other. Corrections are welcome.

*Data and code.* All data, code and derivation scripts are at `https://github.com/nmoinvaz/choptuik-large-d`, with a reproduction script (`./reproduce.sh`); the double-double points are in `hp/data/`, the matter computations in `solver/matter/`.

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